Data Flow Diagram

Date	15 March 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

OptiCrop - Data Flow Diagram

1. External Entity (Farmer): enters N, P, K, temperature, humidity, pH and rainfall via the web form.
2. Process 1 - Input Validation: the Flask backend parses and validates the 7 input parameters.
3. Process 2 - Prediction: the inputs form a NumPy array passed to the trained Logistic Regression model.
4. Data Store: pre-trained model weights (models/model.pkl) and the training dataset (Crop_recommendation.csv).
5. Process 3 - Result: the predicted optimal crop is generated.
6. External Entity (Farmer): the recommended crop is returned on the HTML result page.